

MPG No. 3.3.8 2' [Type III, monoclinic] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} M_x & 0 & M_z \end{bmatrix}$$

$$M_x = M_{px}^{(1)}$$

$$M_z = M_{pz}^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ M_{xy} & 0 & M_{yz} \\ 0 & M_{yz} & 0 \end{bmatrix}$$

$$M_{xy} = M_{dxy}^{(1)}$$

$$M_{yz} = M_{dyz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & M_{yz} \\ 0 & -M_{yz} & 0 \end{bmatrix}$$

$$M_{xy} = T_{pz}^{(1)}$$

$$M_{yz} = T_{px}^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & 0 & M_{xxz} \\ M_{yyx} & 0 & M_{yyz} \\ M_{zzx} & 0 & M_{zzz} \\ 0 & M_{yyz} & 0 \\ M_{xxz} & 0 & M_{zzx} \\ 0 & M_{yyx} & 0 \end{bmatrix}$$

$$M_{xxx} = 2M_{fax}^{(1)} + 3M_{px}^{(1)}$$

$$M_{xxz} = -M_{faz}^{(1)} + M_{fbz}^{(1)} + M_{pz}^{(1)}$$

$$M_{yyx} = -M_{fax}^{(1)} + M_{fbx}^{(1)} + M_{px}^{(1)}$$

$$M_{yyz} = -M_{faz}^{(1)} - M_{fbz}^{(1)} + M_{pz}^{(1)}$$

$$M_{zzx} = -M_{fax}^{(1)} - M_{fbx}^{(1)} + M_{px}^{(1)}$$

$$M_{zzz} = 2M_{faz}^{(1)} + 3M_{pz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ M_{yyx} & 0 & M_{yyz} \\ M_{zzx} & 0 & 0 \\ 0 & -\frac{M_{yyz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & 0 & -\frac{M_{zzx}}{2} \\ 0 & -\frac{M_{yyx}}{2} & 0 \end{bmatrix}$$

$$M_{xxz} = 2M_{pz}^{(2)} + 2T_{dxy}^{(1)}$$

$$M_{yyx} = 2M_{px}^{(2)} + 2T_{dyz}^{(1)}$$

$$M_{yyz} = 2M_{pz}^{(2)} - 2T_{dxy}^{(1)}$$

$$M_{zzx} = 2M_{px}^{(2)} - 2T_{dyz}^{(1)}$$

$$\frac{M_{yyz}}{2} = M_{pz}^{(2)} - T_{dxy}^{(1)}$$

$$\frac{M_{xxz}}{2} = M_{pz}^{(2)} + T_{dxy}^{(1)}$$

$$\frac{M_{zzx}}{2} = M_{px}^{(2)} - T_{dyz}^{(1)}$$

$$\frac{M_{yyx}}{2} = M_{px}^{(2)} + T_{dyz}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & M_{yzy} & 0 \\ M_{zxx} & 0 & M_{zxx} \\ 0 & M_{xyy} & 0 \end{bmatrix}$$

$$M_{yzy} = M_{pz}^{(3)} + T_{dxy}^{(2)}$$

$$M_{zxx} = -M_{pz}^{(3)} + T_{dxy}^{(2)}$$

$$M_{zzx} = M_{px}^{(3)} + T_{dyz}^{(2)}$$

$$M_{xyy} = -M_{px}^{(3)} + T_{dyz}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & M_{xxxy} \\ 0 & 0 & 0 & M_{yyyz} & 0 & M_{yyxy} \\ 0 & 0 & 0 & M_{zzyz} & 0 & M_{zzxy} \\ M_{txyz} & M_{yyyz} & M_{zzyz} & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & M_{zxyx} \\ M_{xxxy} & M_{yyxy} & M_{zzxy} & 0 & M_{zxyx} & 0 \end{bmatrix}$$

$$M_{xxyz} = M_{dyz}^{(1)} + 2M_{gbx}^{(1)}$$

$$M_{xxxy} = M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{gaz}^{(1)} - M_{gbz}^{(1)}$$

$$M_{yyyz} = M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{gax}^{(1)} - M_{gbx}^{(1)}$$

$$M_{yyxy} = M_{dxy}^{(1)} + 2M_{dxy}^{(2)} - M_{gaz}^{(1)} - M_{gbz}^{(1)}$$

$$M_{zzyz} = M_{dyz}^{(1)} + 2M_{dyz}^{(2)} - M_{gax}^{(1)} - M_{gbx}^{(1)}$$

$$M_{zzxy} = M_{dxy}^{(1)} + 2M_{gbz}^{(1)}$$

$$M_{yzzx} = M_{dxy}^{(2)} + 2M_{gbz}^{(1)}$$

$$M_{zxyx} = M_{dyz}^{(2)} + 2M_{gbx}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & M_{xxxy} \\ 0 & 0 & 0 & M_{yyyz} & 0 & M_{yyxy} \\ 0 & 0 & 0 & M_{zzyz} & 0 & M_{zzxy} \\ -M_{xxyz} & -M_{yyyz} & -M_{zzyz} & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & M_{zxyx} \\ -M_{xxxy} & -M_{yyxy} & -M_{zzxy} & 0 & -M_{zxyx} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxyz} &= M_{dyz}^{(3)} - 4T_{fbx}^{(1)} \\
M_{xxxy} &= M_{dxy}^{(3)} - 2T_{faz}^{(1)} + 2T_{fbz}^{(1)} + 2T_{pz}^{(1)} \\
M_{yyyy} &= M_{dyz}^{(3)} - 2T_{faz}^{(1)} + 2T_{fbx}^{(1)} + 2T_{px}^{(1)} \\
M_{yyxy} &= M_{dxy}^{(3)} + 2T_{faz}^{(1)} + 2T_{fbz}^{(1)} - 2T_{pz}^{(1)} \\
M_{zzyz} &= M_{dyz}^{(3)} + 2T_{fax}^{(1)} + 2T_{fbx}^{(1)} - 2T_{px}^{(1)} \\
M_{zzxy} &= M_{dxy}^{(3)} - 4T_{fbz}^{(1)} \\
M_{yzzx} &= -4T_{faz}^{(1)} - T_{pz}^{(1)} \\
M_{zxxxy} &= -4T_{fax}^{(1)} - T_{px}^{(1)}
\end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & M_{zxxxy} \\ 0 & M_{zxxxy} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzzx} &= 2M_{dxy}^{(4)} \\
M_{zxxxy} &= 2M_{dyz}^{(4)}
\end{aligned}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & M_{zxxxy} \\ 0 & -M_{zxxxy} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzzx} &= -T_{pz}^{(2)} \\
M_{zxxxy} &= -T_{px}^{(2)}
\end{aligned}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xxyz} & 0 & M_{xxyy} \\ M_{yyyz} & 0 & M_{yyxy} \\ M_{zzyz} & 0 & M_{zzxy} \\ 0 & M_{yzzx} & 0 \\ M_{zxyz} & 0 & M_{zxxxy} \\ 0 & M_{xyzx} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxyz} &= 2T_{fax}^{(2)} + T_{px}^{(3)} + 2T_{px}^{(4)} \\
M_{xxxy} &= -2M_{dxy}^{(5)} - T_{faz}^{(2)} + T_{fbz}^{(2)} + T_{pz}^{(3)} \\
M_{yyyy} &= -2M_{dyz}^{(5)} - T_{faz}^{(2)} + T_{fbx}^{(2)} + T_{px}^{(3)} \\
M_{yyxy} &= 2M_{dxy}^{(5)} - T_{faz}^{(2)} - T_{fbz}^{(2)} + T_{pz}^{(3)} \\
M_{zzyz} &= 2M_{dyz}^{(5)} - T_{fax}^{(2)} - T_{fbx}^{(2)} + T_{px}^{(3)} \\
M_{zzxy} &= 2T_{faz}^{(2)} + T_{pz}^{(3)} + 2T_{pz}^{(4)} \\
M_{yzzx} &= -M_{dxy}^{(5)} - T_{faz}^{(2)} - T_{fbz}^{(2)} + T_{pz}^{(4)} \\
M_{zxyz} &= M_{dxy}^{(5)} - T_{faz}^{(2)} + T_{fbz}^{(2)} + T_{pz}^{(4)} \\
M_{zxxxy} &= -M_{dyz}^{(5)} - T_{fax}^{(2)} - T_{fbx}^{(2)} + T_{px}^{(4)} \\
M_{xyzx} &= M_{dyz}^{(5)} - T_{fax}^{(2)} + T_{fbx}^{(2)} + T_{px}^{(4)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{zxyz} & 0 & M_{zxxxy} \\ M_{xyxx} & M_{xyyy} & M_{xyzz} & 0 & M_{xyzx} & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yzxx} &= 2T_{fax}^{(3)} + T_{px}^{(5)} + 2T_{px}^{(6)} \\
M_{yzyy} &= -2M_{dyz}^{(6)} - T_{fax}^{(3)} + T_{fbx}^{(3)} + T_{px}^{(5)} \\
M_{yzzz} &= 2M_{dyz}^{(6)} - T_{fax}^{(3)} - T_{fbx}^{(3)} + T_{px}^{(5)} \\
M_{yzzx} &= M_{dxy}^{(6)} - T_{fax}^{(3)} + T_{fbz}^{(3)} + T_{pz}^{(6)} \\
M_{zxzy} &= -M_{dxy}^{(6)} - T_{faz}^{(3)} - T_{fbz}^{(3)} + T_{pz}^{(6)} \\
M_{zxxxy} &= M_{dyz}^{(6)} - T_{fax}^{(3)} + T_{fbx}^{(3)} + T_{px}^{(6)} \\
M_{xyxx} &= -2M_{dxy}^{(6)} - T_{faz}^{(3)} + T_{fbz}^{(3)} + T_{pz}^{(5)} \\
M_{xyyy} &= 2M_{dxy}^{(6)} - T_{faz}^{(3)} - T_{fbz}^{(3)} + T_{pz}^{(5)} \\
M_{xyzx} &= 2T_{faz}^{(3)} + T_{pz}^{(5)} + 2T_{pz}^{(6)} \\
M_{xyzx} &= -M_{dyz}^{(6)} - T_{fax}^{(3)} - T_{fbx}^{(3)} + T_{px}^{(6)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix}
0 & 0 & 0 & M_{xxyz} & 0 & M_{xxxy} & M_{xxzy} & 0 & M_{xxyx} \\
0 & 0 & 0 & M_{yyyz} & 0 & M_{yyxy} & M_{yyzy} & 0 & M_{yyyx} \\
0 & 0 & 0 & M_{zzyz} & 0 & M_{zzxy} & M_{zzzy} & 0 & M_{zzyx} \\
M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\
0 & 0 & 0 & M_{zxyz} & 0 & M_{zxxxy} & M_{zxzy} & 0 & M_{zxyx} \\
M_{xyxx} & M_{xyyy} & M_{xyzx} & 0 & M_{xyzx} & 0 & 0 & M_{xyxz} & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{xxyz} &= M_{dyz}^{(1)} + M_{dyz}^{(3)} + 2M_{gbx}^{(1)} + 2T_{fax}^{(2)} - 4T_{fbx}^{(1)} + T_{px}^{(3)} + 2T_{px}^{(4)} \\
M_{xxxy} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{dxy}^{(3)} - 2M_{dxy}^{(5)} + M_{gax}^{(1)} - M_{gbz}^{(1)} - 2T_{faz}^{(1)} - T_{faz}^{(2)} + 2T_{fbz}^{(1)} + T_{fbz}^{(2)} + 2T_{pz}^{(1)} + T_{pz}^{(3)} \\
M_{xxzy} &= M_{dyz}^{(1)} + M_{dyz}^{(3)} + 2M_{gbx}^{(1)} - 2T_{fax}^{(2)} - 4T_{fbx}^{(1)} - T_{px}^{(3)} - 2T_{px}^{(4)} \\
M_{xxyx} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{dxy}^{(3)} + 2M_{dxy}^{(5)} + M_{gax}^{(1)} - M_{gbz}^{(1)} - 2T_{faz}^{(1)} + T_{faz}^{(2)} + 2T_{fbz}^{(1)} - T_{fbz}^{(2)} + 2T_{pz}^{(1)} - T_{pz}^{(3)} \\
M_{yyyz} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} - 2M_{dyz}^{(5)} + M_{gax}^{(1)} - M_{gbx}^{(1)} - 2T_{fax}^{(1)} - T_{fax}^{(2)} + 2T_{fbx}^{(1)} + T_{fbx}^{(2)} + 2T_{px}^{(1)} + T_{px}^{(3)} \\
M_{yyxy} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{dxy}^{(3)} + 2M_{dxy}^{(5)} - M_{gax}^{(1)} - M_{gbz}^{(1)} + 2T_{faz}^{(1)} - T_{faz}^{(2)} + 2T_{fbz}^{(1)} - T_{fbz}^{(2)} - 2T_{pz}^{(1)} + T_{pz}^{(3)} \\
M_{yyzy} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} + 2M_{dyz}^{(5)} + M_{gax}^{(1)} - M_{gbx}^{(1)} - 2T_{fax}^{(1)} + T_{fax}^{(2)} + 2T_{fbx}^{(1)} - T_{fbx}^{(2)} + 2T_{px}^{(1)} - T_{px}^{(3)} \\
M_{yyyx} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{dxy}^{(3)} - 2M_{dxy}^{(5)} - M_{gax}^{(1)} - M_{gbz}^{(1)} + 2T_{faz}^{(1)} + T_{faz}^{(2)} + 2T_{fbz}^{(1)} + T_{fbz}^{(2)} - 2T_{pz}^{(1)} - T_{pz}^{(3)} \\
M_{zzyz} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} + 2M_{dyz}^{(5)} - M_{gax}^{(1)} - M_{gbx}^{(1)} + 2T_{fax}^{(1)} - T_{fax}^{(2)} + 2T_{fbx}^{(1)} - T_{fbx}^{(2)} - 2T_{px}^{(1)} + T_{px}^{(3)} \\
M_{zzxy} &= M_{dxy}^{(1)} + M_{dxy}^{(3)} + 2M_{gbz}^{(1)} + 2T_{faz}^{(2)} - 4T_{fbz}^{(1)} + T_{pz}^{(3)} + 2T_{pz}^{(4)} \\
M_{zzzy} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} - 2M_{dyz}^{(5)} - M_{gax}^{(1)} - M_{gbx}^{(1)} + 2T_{faz}^{(1)} + T_{faz}^{(2)} + 2T_{fbx}^{(1)} + T_{fbx}^{(2)} - 2T_{px}^{(1)} - T_{px}^{(3)} \\
M_{zzyx} &= M_{dxy}^{(1)} + M_{dxy}^{(3)} + 2M_{gbz}^{(1)} - 2T_{faz}^{(2)} - 4T_{fbz}^{(1)} - T_{pz}^{(3)} - 2T_{pz}^{(4)} \\
M_{yzxx} &= M_{dyz}^{(1)} - M_{dyz}^{(3)} + 2M_{gbx}^{(1)} + 4T_{fbx}^{(1)} \\
M_{yzyy} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} - M_{dyz}^{(3)} + M_{gax}^{(1)} - M_{gbx}^{(1)} + 2T_{fax}^{(1)} - 2T_{fbx}^{(1)} - 2T_{px}^{(1)} \\
M_{yzzz} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} - M_{dyz}^{(3)} - M_{gax}^{(1)} - M_{gbx}^{(1)} - 2T_{fax}^{(1)} - 2T_{fbx}^{(1)} + 2T_{px}^{(1)} \\
M_{yzzx} &= M_{dxy}^{(2)} - M_{dxy}^{(5)} + 2M_{gbz}^{(1)} - 4T_{faz}^{(1)} - T_{faz}^{(2)} - T_{fbz}^{(2)} - T_{pz}^{(1)} + T_{pz}^{(4)} \\
M_{yzxz} &= M_{dxy}^{(2)} + M_{dxy}^{(5)} + 2M_{gbz}^{(1)} - 4T_{faz}^{(1)} + T_{faz}^{(2)} + T_{fbz}^{(2)} - T_{pz}^{(1)} - T_{pz}^{(4)} \\
M_{zxyz} &= M_{dxy}^{(2)} + M_{dxy}^{(5)} + 2M_{gbz}^{(1)} + 4T_{faz}^{(1)} - T_{faz}^{(2)} + T_{fbz}^{(2)} + T_{pz}^{(1)} + T_{pz}^{(4)} \\
M_{zxxy} &= M_{dxy}^{(2)} - M_{dxy}^{(5)} + 2M_{gbx}^{(1)} - 4T_{fax}^{(1)} - T_{fax}^{(2)} - T_{fbx}^{(2)} - T_{px}^{(1)} + T_{px}^{(4)} \\
M_{zxyy} &= M_{dxy}^{(2)} - M_{dxy}^{(5)} + 2M_{gbz}^{(1)} + 4T_{faz}^{(1)} + T_{faz}^{(2)} - T_{fbz}^{(2)} + T_{pz}^{(1)} - T_{pz}^{(4)} \\
M_{zxyx} &= M_{dyz}^{(2)} + M_{dyz}^{(5)} + 2M_{gbx}^{(1)} - 4T_{fax}^{(1)} + T_{fax}^{(2)} + T_{fbx}^{(2)} - T_{px}^{(1)} - T_{px}^{(4)} \\
M_{xyxx} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} - M_{dxy}^{(3)} + M_{gax}^{(1)} - M_{gbz}^{(1)} + 2T_{faz}^{(1)} - 2T_{fbz}^{(1)} - 2T_{pz}^{(1)} \\
M_{xyyy} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} - M_{dxy}^{(3)} - M_{gax}^{(1)} - M_{gbz}^{(1)} - 2T_{faz}^{(1)} - 2T_{fbz}^{(1)} + 2T_{pz}^{(1)} \\
M_{xyyz} &= M_{dxy}^{(1)} - M_{dxy}^{(3)} + 2M_{gbz}^{(1)} + 4T_{fbz}^{(1)} \\
M_{xyzx} &= M_{dyz}^{(2)} + M_{dyz}^{(5)} + 2M_{gbx}^{(1)} + 4T_{fax}^{(1)} - T_{fax}^{(2)} + T_{fbx}^{(2)} + T_{px}^{(1)} + T_{px}^{(4)} \\
M_{xyxz} &= M_{dyz}^{(2)} - M_{dyz}^{(5)} + 2M_{gbx}^{(1)} + 4T_{fax}^{(1)} + T_{fax}^{(2)} - T_{fbx}^{(2)} + T_{px}^{(1)} - T_{px}^{(4)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix}
M_{yzxx} & M_{yzyy} & M_{yzzz} & 0 & M_{yzxz} & 0 & 0 & M_{yzxz} & 0 \\
0 & 0 & 0 & M_{zxyz} & 0 & M_{zxxy} & M_{zxyz} & 0 & M_{zxyx} \\
M_{xyxx} & M_{xyyy} & M_{xyyz} & 0 & M_{xyzx} & 0 & 0 & M_{xyxz} & 0
\end{bmatrix}$$

$$\begin{aligned}
M_{yzxx} &= 2T_{fax}^{(3)} + T_{px}^{(5)} + 2T_{px}^{(6)} \\
M_{yzyy} &= -2M_{dyz}^{(6)} - T_{fax}^{(3)} + T_{fbx}^{(3)} + T_{px}^{(5)} \\
M_{yzzz} &= 2M_{dyz}^{(6)} - T_{fax}^{(3)} - T_{fbx}^{(3)} + T_{px}^{(5)} \\
M_{yzxz} &= 2M_{dxy}^{(4)} + M_{dxy}^{(6)} - T_{faz}^{(3)} + T_{fbz}^{(3)} - T_{pz}^{(2)} + T_{pz}^{(6)} \\
M_{yzzx} &= -2M_{dxy}^{(4)} + M_{dxy}^{(6)} - T_{faz}^{(3)} + T_{fbz}^{(3)} + T_{pz}^{(2)} + T_{pz}^{(6)} \\
M_{zxyz} &= 2M_{dxy}^{(4)} - M_{dxy}^{(6)} - T_{faz}^{(3)} - T_{fbz}^{(3)} + T_{pz}^{(2)} + T_{pz}^{(6)} \\
M_{zxxy} &= 2M_{dyz}^{(4)} + M_{dyz}^{(6)} - T_{fax}^{(3)} + T_{fbx}^{(3)} - T_{px}^{(2)} + T_{px}^{(6)} \\
M_{zxxz} &= -2M_{dxy}^{(4)} - M_{dxy}^{(6)} - T_{faz}^{(3)} - T_{fbz}^{(3)} - T_{pz}^{(2)} + T_{pz}^{(6)} \\
M_{zxyx} &= -2M_{dyz}^{(4)} + M_{dyz}^{(6)} - T_{fax}^{(3)} + T_{fbx}^{(3)} + T_{px}^{(2)} + T_{px}^{(6)} \\
M_{xyxx} &= -2M_{dxy}^{(6)} - T_{faz}^{(3)} + T_{fbz}^{(3)} + T_{pz}^{(5)} \\
M_{xyyy} &= 2M_{dxy}^{(6)} - T_{faz}^{(3)} - T_{fbz}^{(3)} + T_{pz}^{(5)} \\
M_{xyzx} &= 2T_{faz}^{(3)} + T_{pz}^{(5)} + 2T_{pz}^{(6)} \\
M_{xyxz} &= 2M_{dyz}^{(4)} - M_{dyz}^{(6)} - T_{faz}^{(3)} - T_{fbx}^{(3)} + T_{px}^{(2)} + T_{px}^{(6)} \\
M_{xyzz} &= -2M_{dyz}^{(4)} - M_{dyz}^{(6)} - T_{fax}^{(3)} - T_{fbx}^{(3)} - T_{px}^{(2)} + T_{px}^{(6)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
0 & M_{xxxy} & 0 \\
M_{yyyyx} & 0 & M_{yyyyz} \\
0 & M_{zzzy} & 0 \\
0 & M_{yyzy} & 0 \\
0 & M_{zzxy} & 0 \\
M_{xxyx} & 0 & M_{xxyz} \\
M_{yzzx} & 0 & M_{yzzz} \\
0 & M_{zxxxy} & 0 \\
0 & M_{xyyy} & 0 \\
-\frac{M_{yyyyz}}{2} + \frac{M_{yyzyz}}{2} + M_{zxxxy} & 0 & \frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} - M_{yzyx} + M_{zzxy}
\end{bmatrix}$$

$$\begin{aligned}
M_{xxxy} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{dxy}^{(3)} - 2M_{dxy}^{(5)} + M_{gax}^{(1)} - M_{gbz}^{(1)} \\
M_{yyyyx} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{dxy}^{(3)} - 2M_{dxy}^{(5)} - M_{gax}^{(1)} - M_{gbz}^{(1)} \\
M_{yyyyz} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} - 2M_{dyz}^{(5)} + M_{gax}^{(1)} - M_{gbx}^{(1)} \\
M_{zzzy} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} - 2M_{dyz}^{(5)} - M_{gax}^{(1)} - M_{gbx}^{(1)} \\
M_{yyzy} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} + M_{dyz}^{(3)} + 2M_{dyz}^{(5)} + M_{gax}^{(1)} - M_{gbx}^{(1)} \\
M_{zzxy} &= M_{dxy}^{(1)} + M_{dxy}^{(3)} + 2M_{gbz}^{(1)} \\
M_{xxyx} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{dxy}^{(3)} + 2M_{dxy}^{(5)} + M_{gax}^{(1)} - M_{gbz}^{(1)} \\
M_{xxyz} &= M_{dyz}^{(1)} + M_{dyz}^{(3)} + 2M_{gbx}^{(1)} \\
M_{yzzx} &= M_{dxy}^{(2)} - M_{dxy}^{(5)} + 2M_{gbz}^{(1)} \\
M_{yzzz} &= M_{dyz}^{(1)} + 2M_{dyz}^{(2)} - M_{dyz}^{(3)} - M_{gax}^{(1)} - M_{gbx}^{(1)} \\
M_{zxxxy} &= M_{dyz}^{(2)} - M_{dyz}^{(5)} + 2M_{gbx}^{(1)} \\
M_{xyyy} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} - M_{dxy}^{(3)} - M_{gax}^{(1)} - M_{gbz}^{(1)}
\end{aligned}$$